Name :- Prabu S
Reg. No :- 192319011

Test

Part-A

1) what is a function? why is it used?

function is an set of instructions that are bundled and can be used in the code wherever we want.

Syntax :-

```
def sum(a,b):
    return a
```

why is it used :

- To stop using reapeated code
- Maximize the optimazation of whole Code

2) Difference between arugument & parameters?

Arguments	Parameters
• Arguments are the code that will work to exeute the program eg:- def sum (a,b): $\boxed{c = a+b}$	Parameters are the data attributes that used to define the program it is defined in parantheses. eg:- def sum $\boxed{(a,b):}$

3) what is return Statement?

A return statement is used in function to call back the Parameters that are executed there are different type of function based on return statement :- without, with statement.

Syntax:

```
def python(a)
    return python
```

4) What is class and object?

- class and object are concepts from OOPS (object oriented programming language).

- where class is blueprint / template that is used to act as framework for program.

- whereas object has the data the attribute that the program used.

5) Define Inheritance?

Inheritance in OOPS is similar to inheritance in human where child inherits from parent class.

There are five types:

- Single
- Multiple
- Multi-level
- Hierchy
- Hybbrid

6) What is method overriding?

Method overriding is used in polymorphism to override an statement without executing to an another one the compile time will be excluded by runtime

7) what is list? Give example?

- List is one form of data structure that python uses to store different variable.

- It is ordered and mutable can be edited, it is denoted by square brackets [ ]

eg:- a = ["Prabu", "BME", 123, "Sarvetha"]

8) Diffance between append () and extend ()?

append ():- append command is mostly used in list add an extra variables to exspiting list

eg:-

```
a = [c,d,e]
a.append (f)
print (a)
```

output:
= [c,d,e,f]

extend ():- extend command is used remove an variable from the list

eg:-

```
a = [c,d,e,f]
a.extend (d)
print(a)
```

output:-
= [c,e,f]

9) what is string slicing?

String slicing is done in python that can used to extract an individual element from the list.

- It has two types positive & negative indexing

Output:- BM

syntax:

```
a = "BME"
print(a[::-1])
```

10) Difference between upper () & lower ?

- Upper() is used to Capitalize the first word of the given string

Output:

Orange

syntax:-

```
a = "orange"
a.upper()
print(a)
```

- Lower() is used to convert lower case of the first element of given string

Output:-

orange

Syntax:-

```
a = "Orange"
a.lower()
print(a)
```

Part - B

1) Write a function to find factorial of a number.

```
a = int (input ("Enter an number :"))
f = 0
for i in range (1, n+1):
    f *= 1
    print (i)
print ("Ur answer:", f)
```

eg:-
input:- Enter an number : 5
Ur answer : 125

2) Write a program to reverse a string.

```
a = input ("Enter an string:")
b = a [::-1]
print ("Reversed string word:", b)
```

eg:-
input:- Enter an string : MAC
Output:- Reversed string word : CAM

3) Write a program to find the largest number in a list

```
a = 3
b = 2
if a > b:
    print ("a is greater than b")
else:
    print ("b is greater than a")
```

Output:

4) Create a class with name & marks, display details

```
class student ( ):
    def __init__ (self, name, marks):
        self.name = name
        self.marks = marks

    def display (self):
        print ("Name:", self.name)
        print ("Mark:", self.marks)

s = students ("Prabu", 100)

s.display
```

Output:-
Prabu
100

5) Write a function to find sum of elements in a list.

```
a = [10, 20, 30]
~~b = [40, 50, 60]~~
b = sum[0] + sum[1] + sum[2]
print(b)
```

~~Output~~:

Output:- 60